

Weather and Climate Big Data

기상기후빅데이터

Weather and Climate Big Data 2

오승민
Spring 2025

Today's Topics

- Characteristics of Weather and Climate Data
- Spatial Data Processing: Interpolation Methods

Weather and Climate Data

- Observations: direct ground-based measurements
- Remote Sensing: Indirect satellite or radar data with global coverage
- Reanalysis: Blended model + observation datasets
- Simulations: Pure model output for testing or forecasting
 - Hindcast: Past model runs for validation against observed data
 - Forecast: Near-future predictions from numerical weather or climate models
 - Paleoclimate (Reconstructed): Ancient climate estimates from proxies

Weather and Climate Data

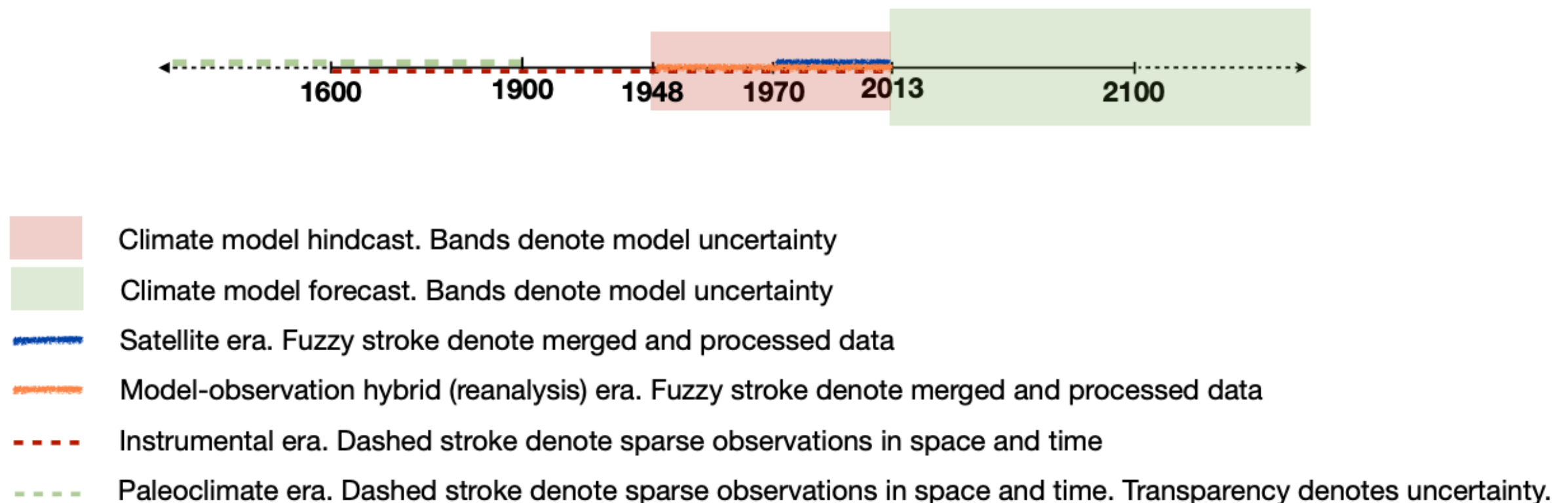


Fig. 1 Climate science has numerous types of data, each with its own challenges.

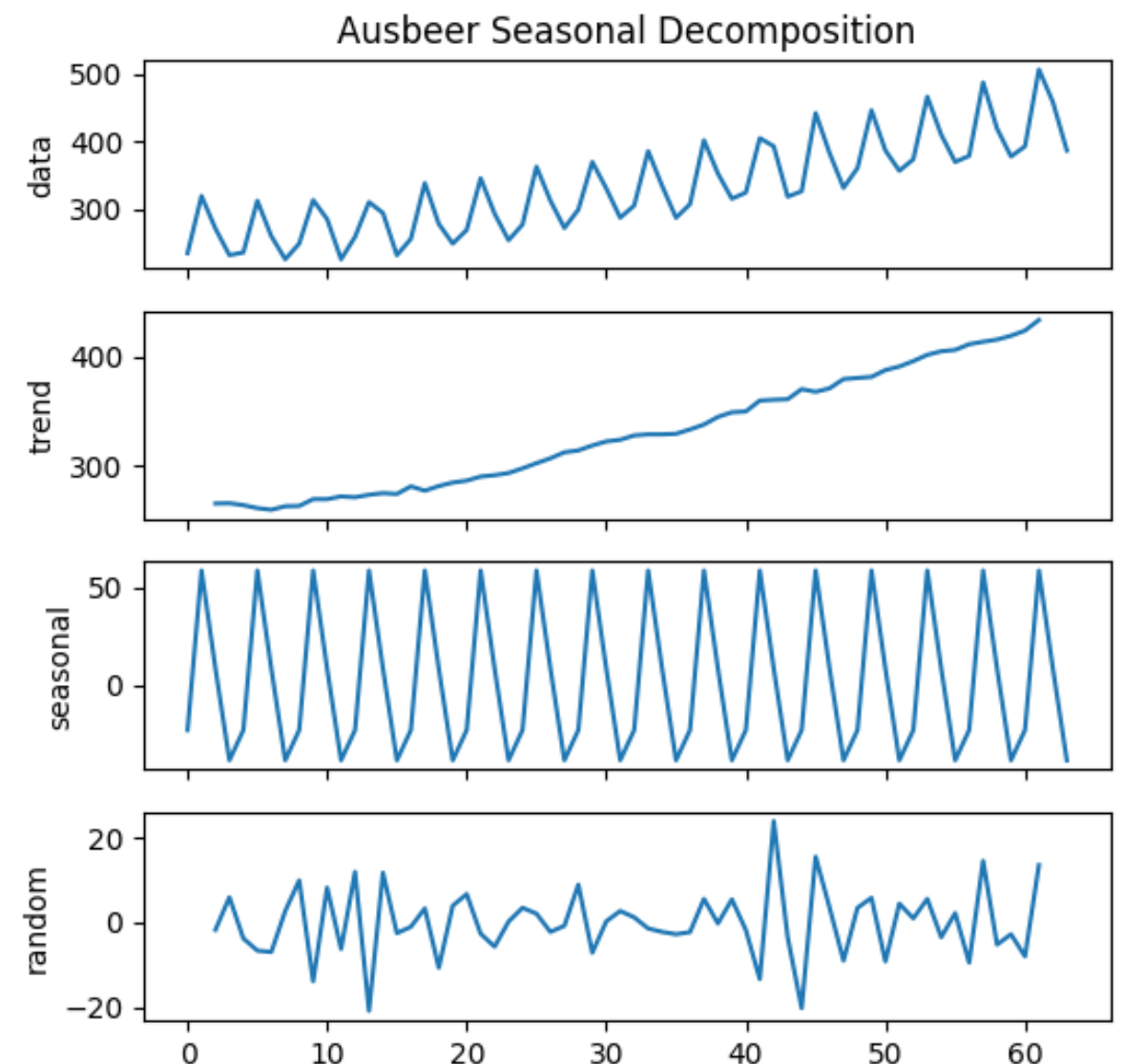
https://climatechange.cs.umn.edu/eddies/media/papers/faghmous_stdm_book_chapter_2013.pdf

Data Characteristics

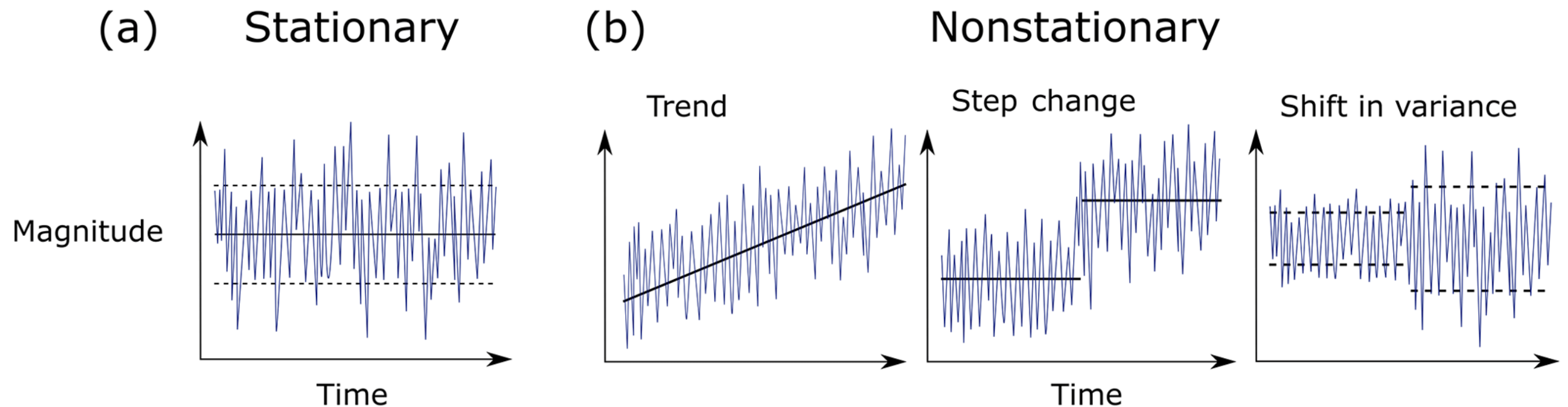
- Large volume and high dimensionality
- Temporal (Time-series) nature and spatial variability
- Multivariate structure
- Missing values and noise
- Non-linearity and non-stationarity
- Multi-scale resolution

Temporal features

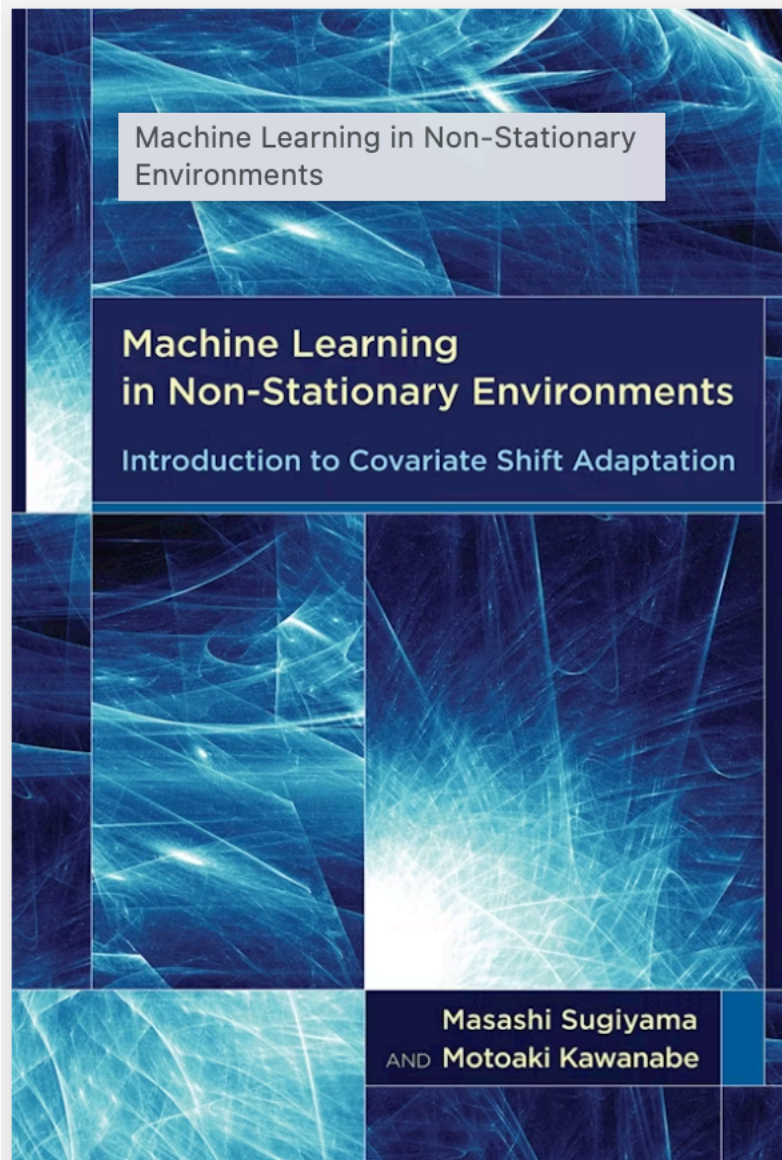
- **Trend:** Long-term increase or decrease over time
- **Seasonality:** Regular, repeating patterns at fixed intervals (e.g., yearly cycles)
- **Cyclic:** Irregular, longer-term oscillations (e.g. El Niño–Southern Oscillation)
- **Residual / Noise:** Random variation not explained by other components



(Non-)Stationarity



(Non-)Stationarity



Adaptive Computation and Machine Learning series

Machine Learning in Non-Stationary Environments

Introduction to Covariate Shift Adaptation

By Masashi Sugiyama and Motoaki Kawanabe

Hardcover

-

\$55.00

Hardcover

ISBN: 9780262017091

Pub date: March 30, 2012

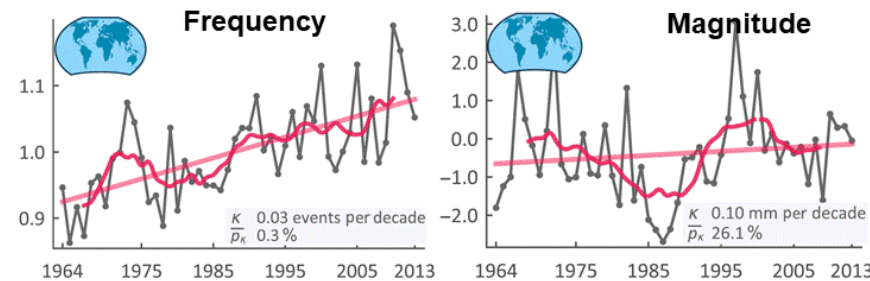
Publisher: The MIT Press

280 pp., 6 x 9 in, 78 b&w illus.

Trend: Underlying Systematic Change

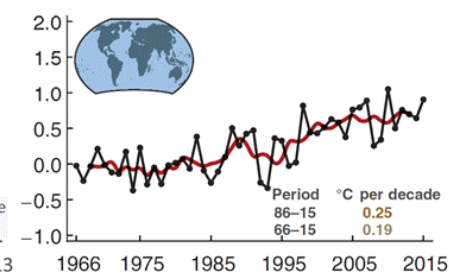
(a) Global extreme daily precipitation

From Papalexiou et al. (2019)



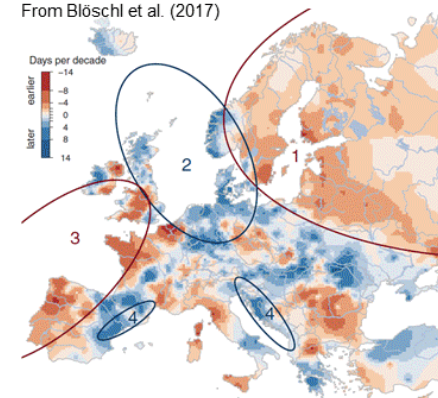
(b) Highest temperature of the year

From Papalexiou et al. (2018)



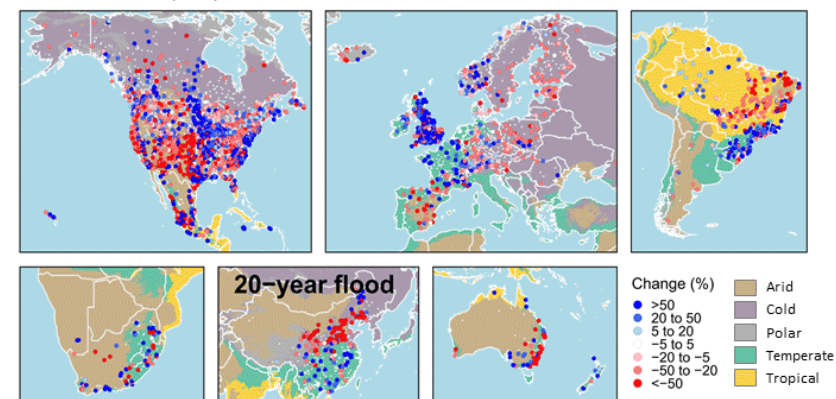
(c) Flood timing

From Blöschl et al. (2017)



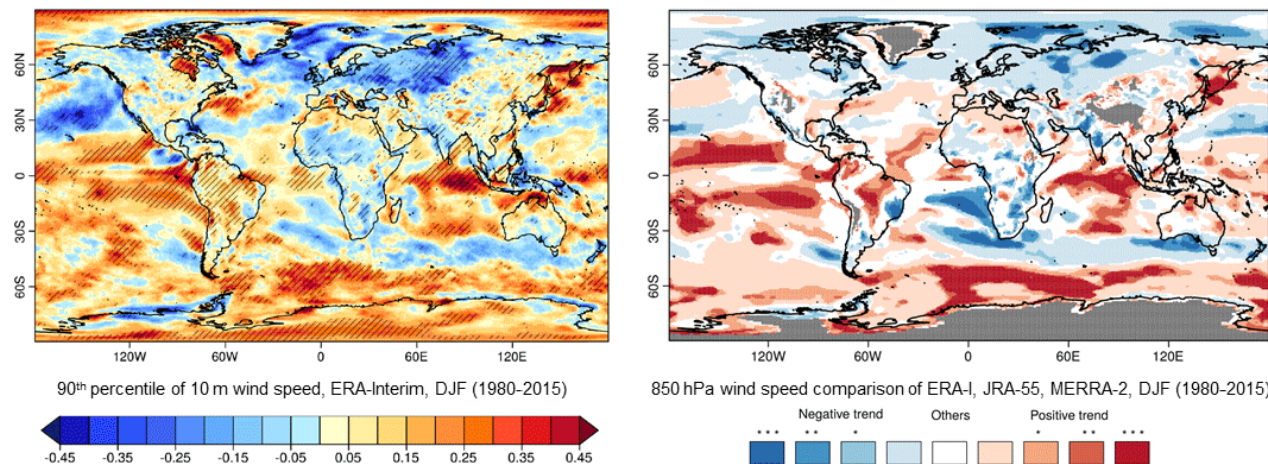
(d) Flood magnitude

From Slater et al. (2021)



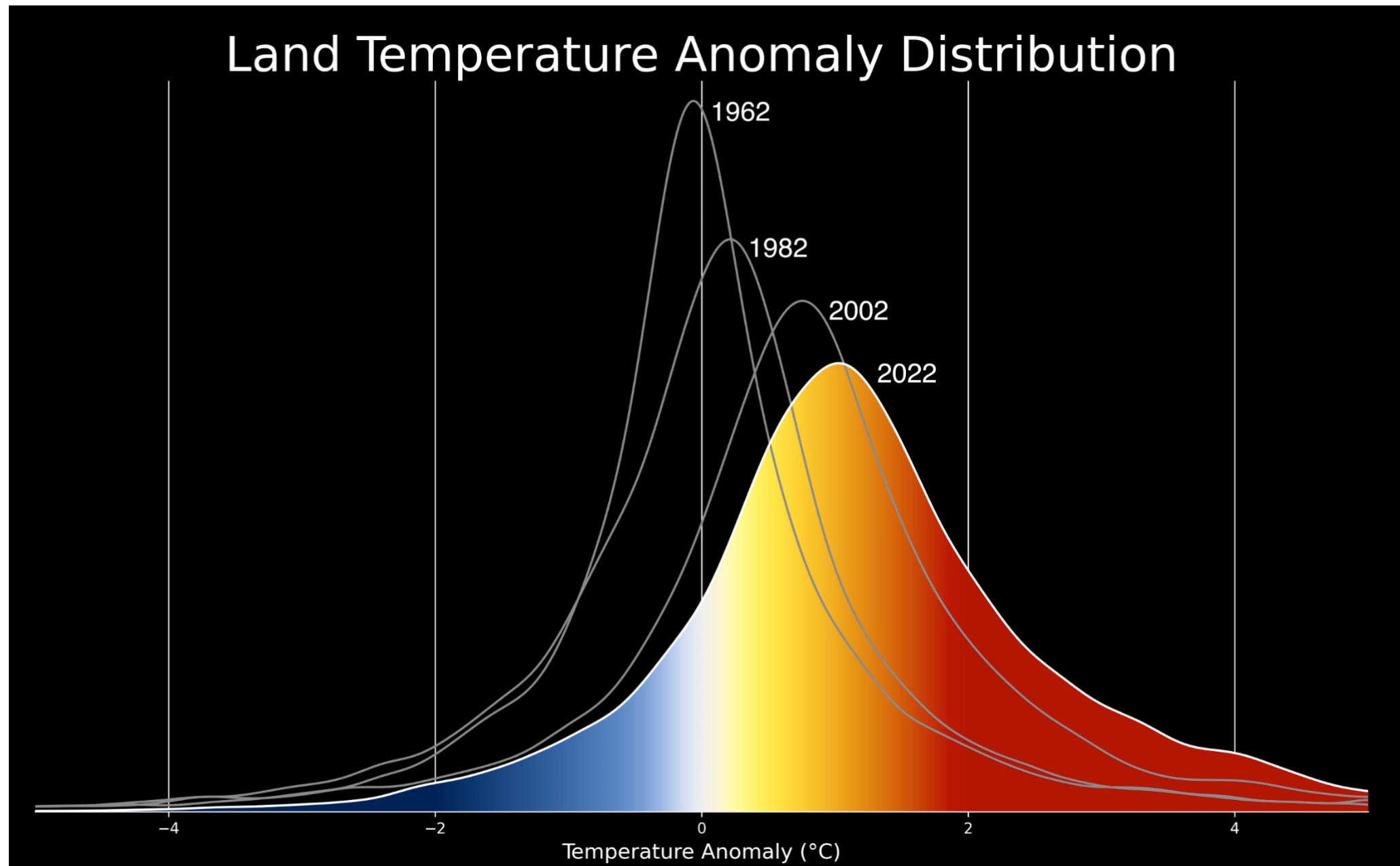
(e) Global wind speed

From Torralba et al. (2017)



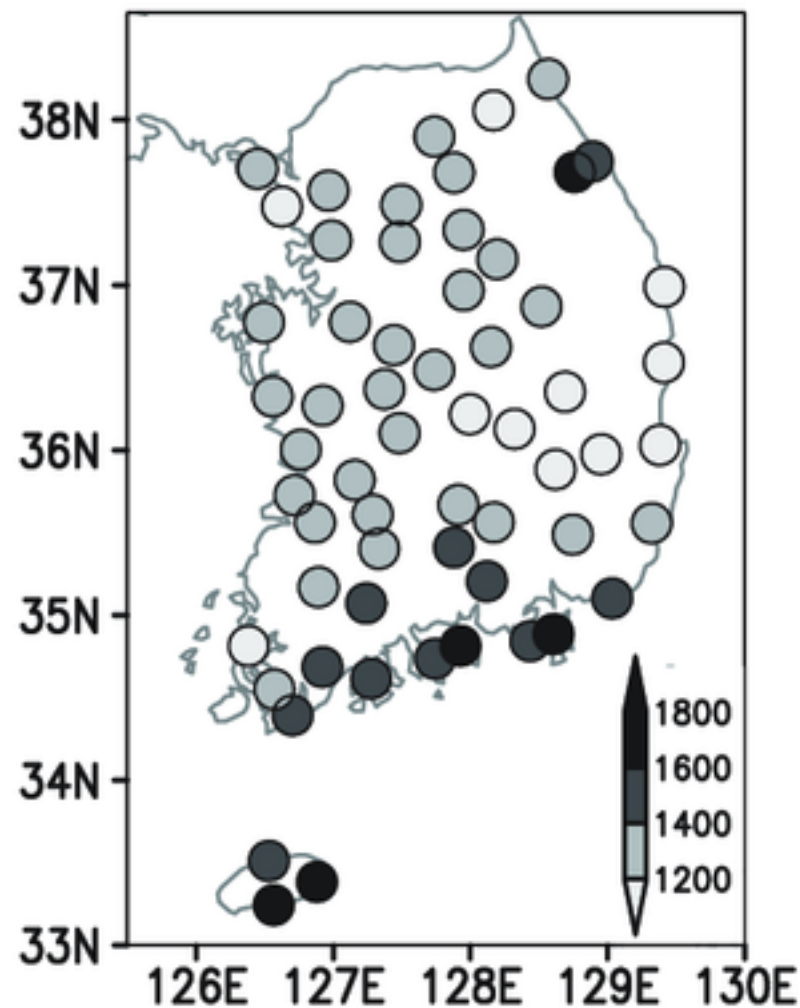
Non-stationarity & Trends

→ More Extremes

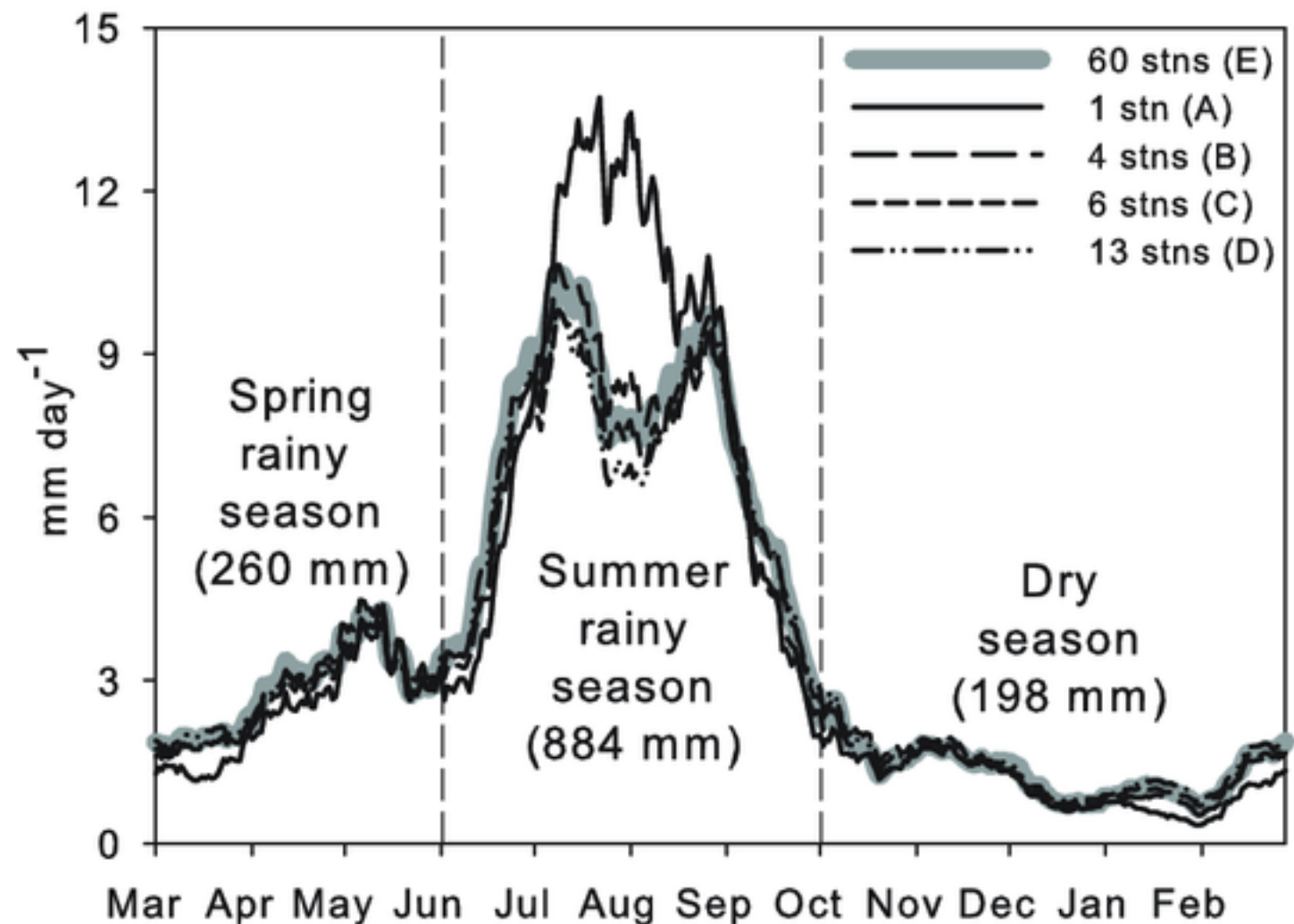


Seasonality in Precipitation

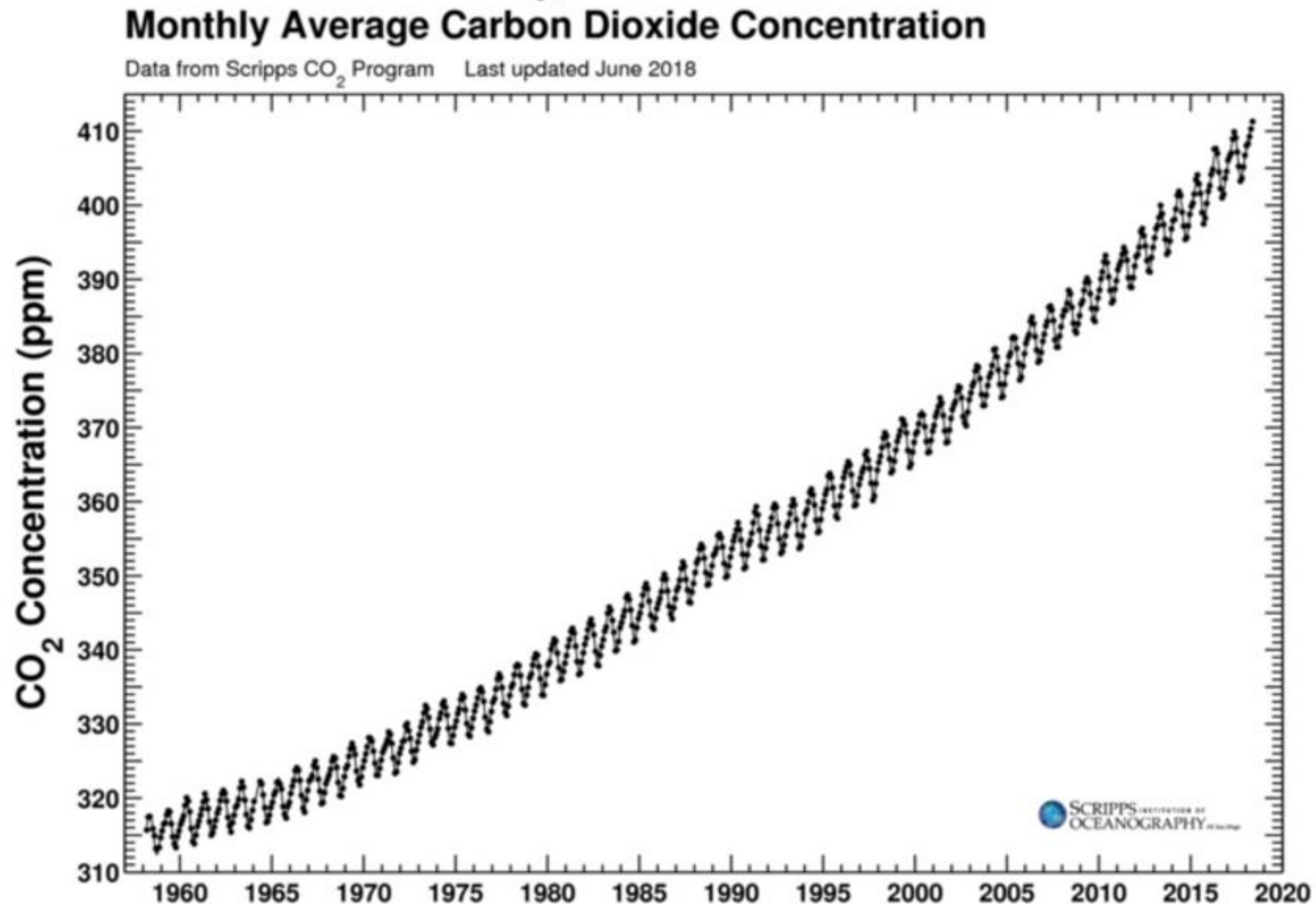
(a) Mean annual precip. (mm)



(b) Mean seasonal precip. cycle

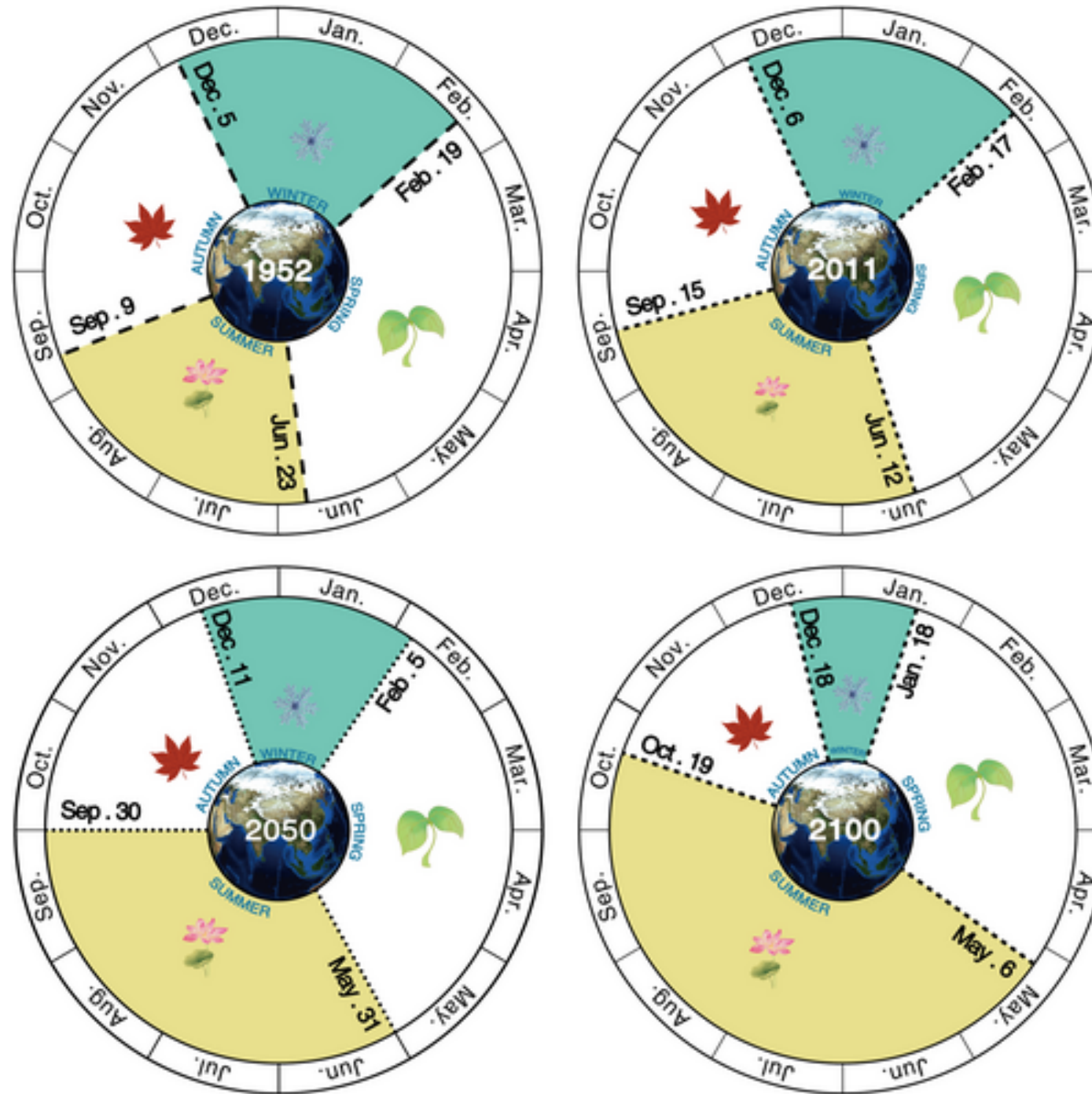


Seasonality in CO₂



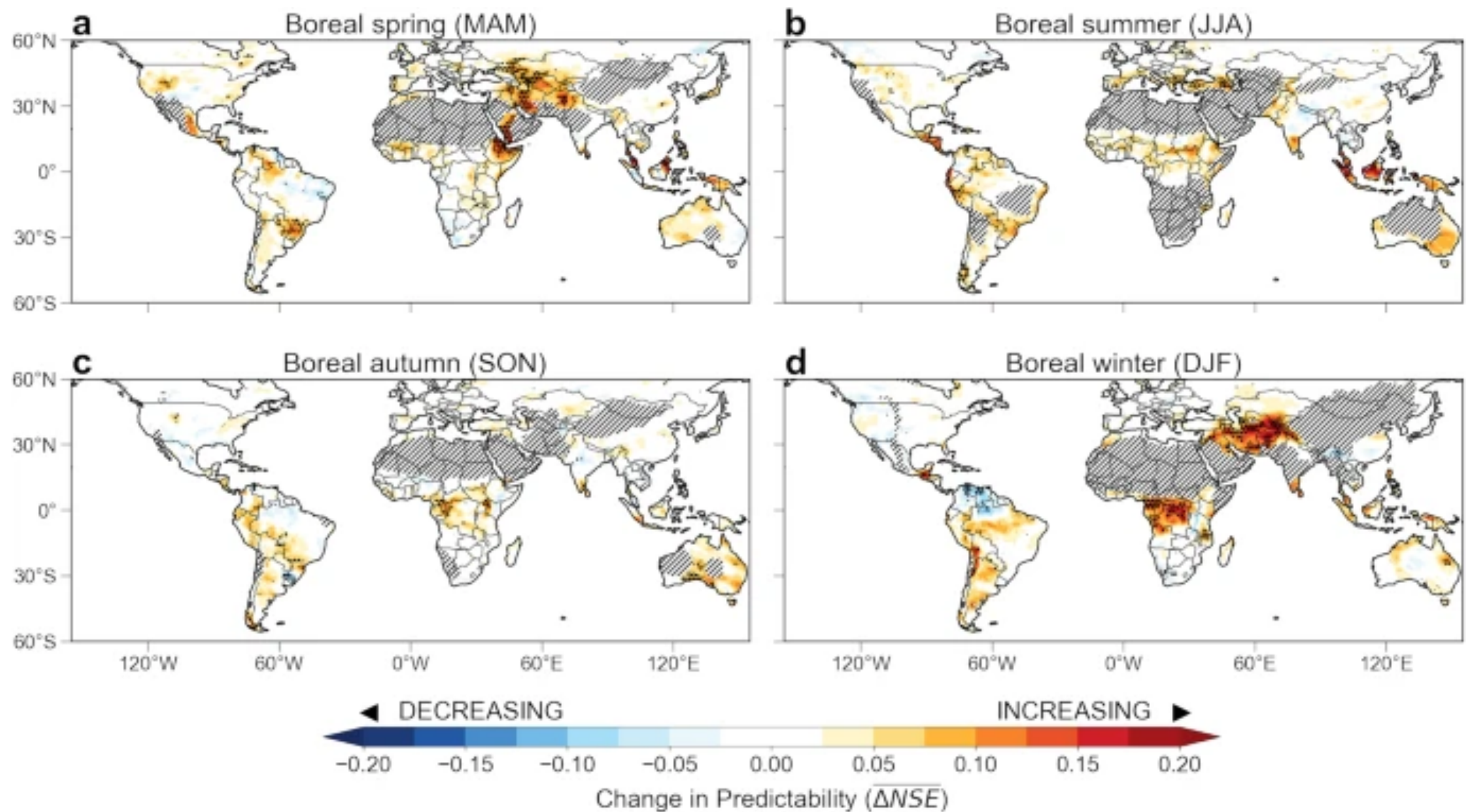
<https://education.nationalgeographic.org/resource/keeling-curve/>

Changes in Seasonality



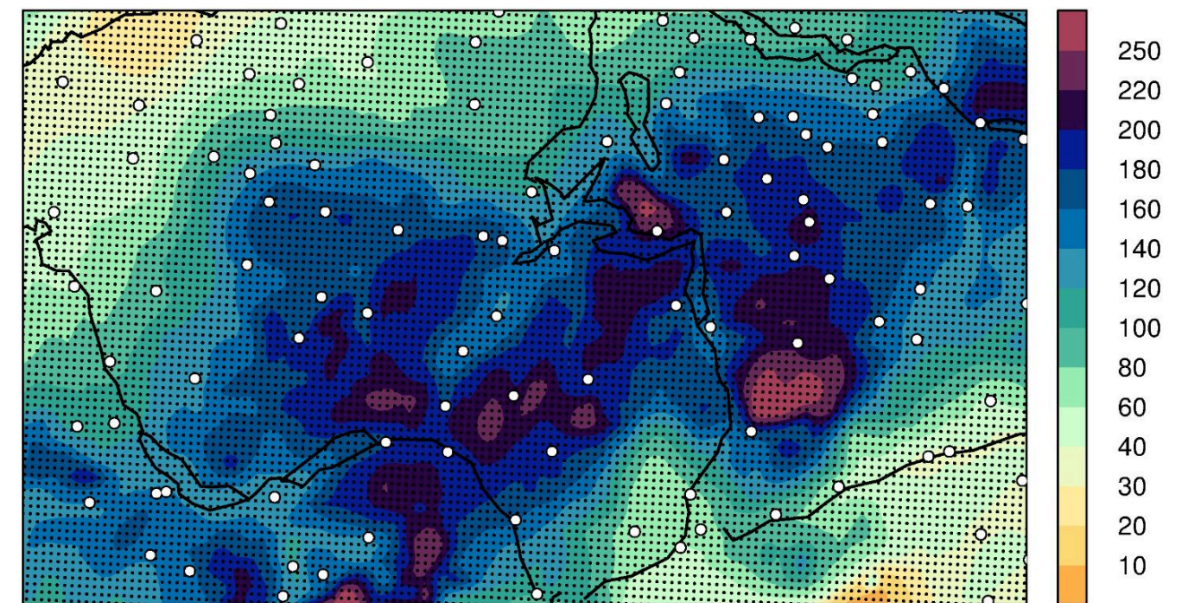
<https://doi.org/10.1029/2020GL091753>

Changes in Seasonality



Spatial features

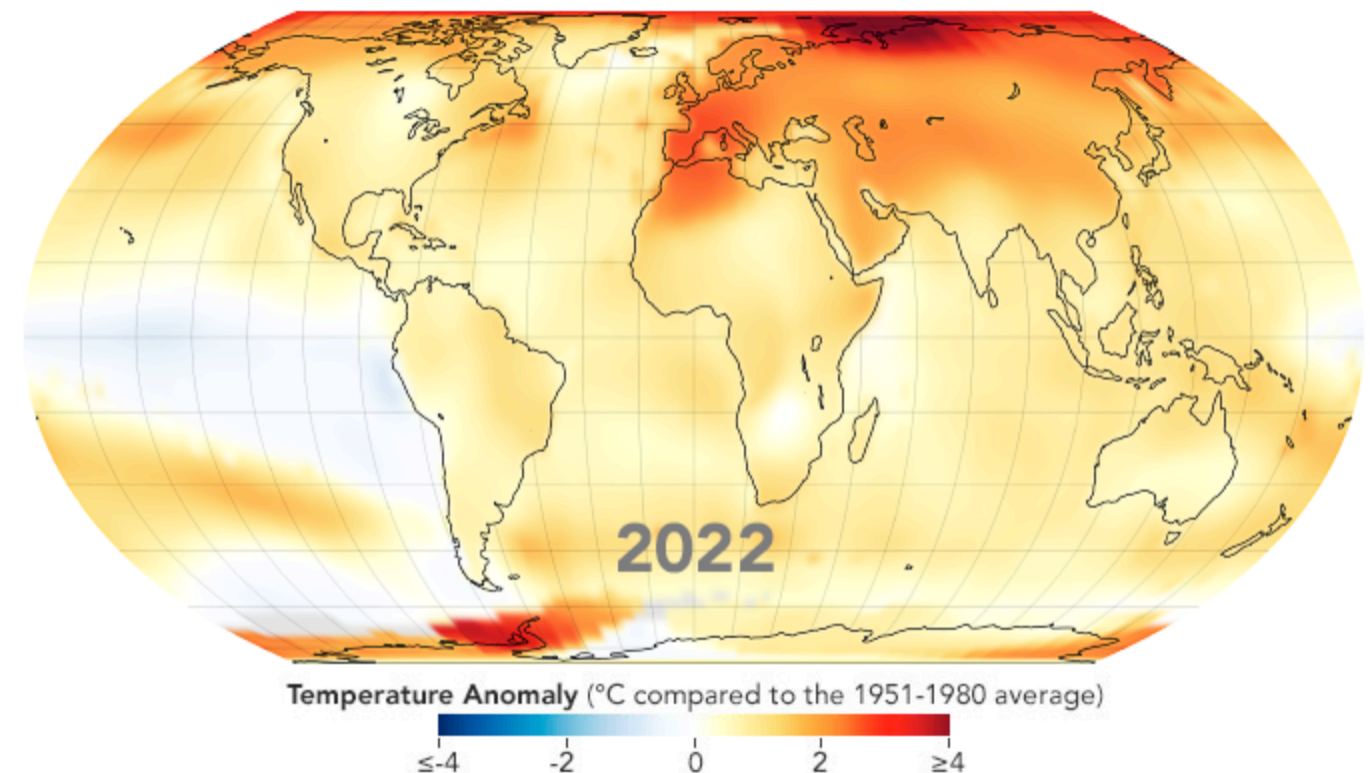
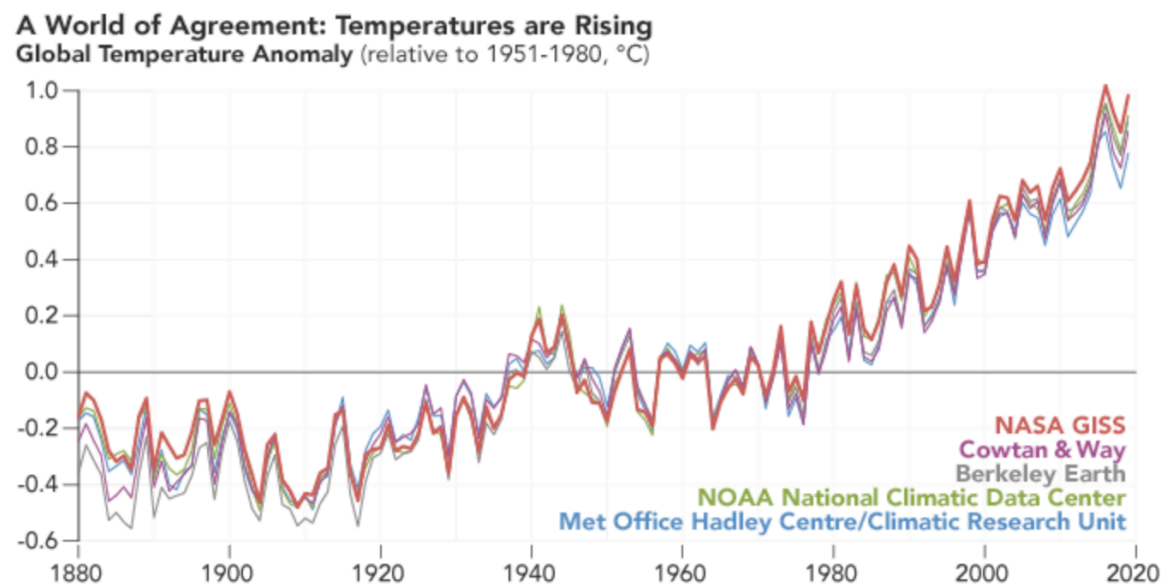
- **Weather and climate data have a spatial dimension:**
Earth system variables (e.g., temperature, precipitation) vary not only over time, but also across space.
- **Data typically come in two forms:** 1) grid-based data from models or satellites and 2) point-based data from weather stations



Example of a spatial analysis of the precipitation (mm) accumulated on August 21st and 22nd , 2005 (section of Switzerland). The analysis uses measurements at 400 stations across the country (white circles). The spacing of the analysis grid points (black dots) is 1 km.

Spatial Trend

- A **spatial trend** refers to the gradual change of climate variables (e.g., temperature, precipitation) over a broad geographic area.

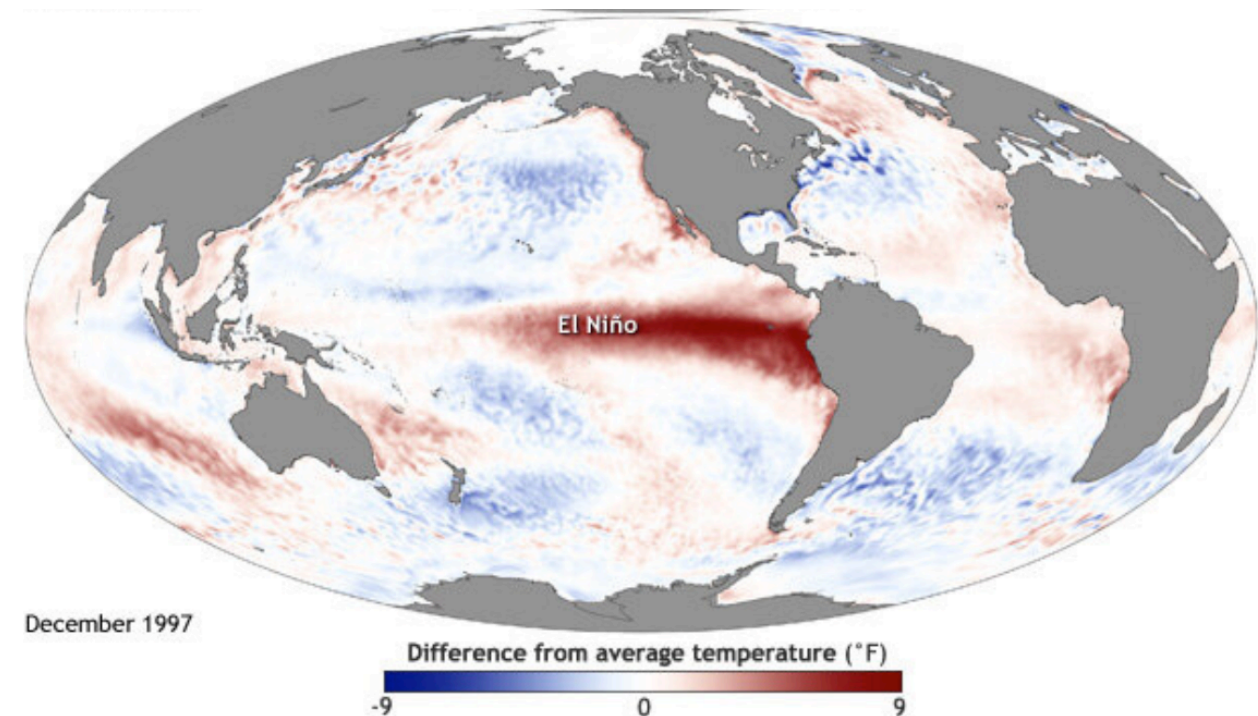
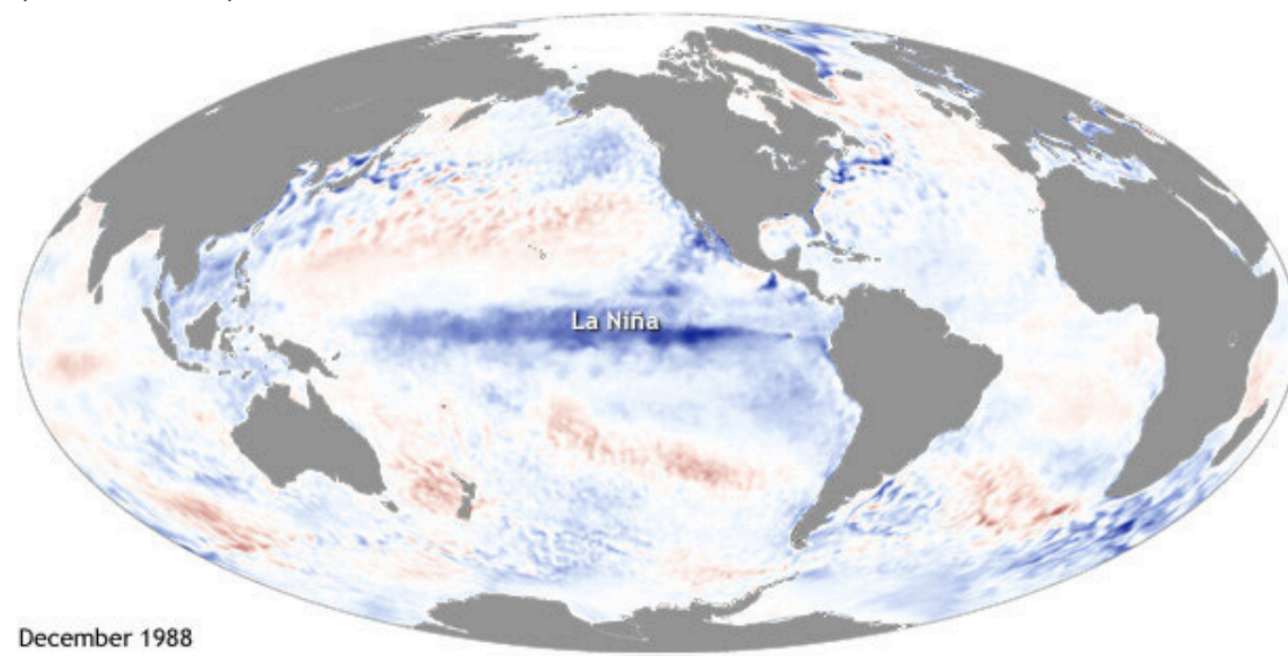


<https://climate.nasa.gov/vital-signs/global-temperature/?intent=121>

<https://earthobservatory.nasa.gov/world-of-change/global-temperatures>

Recurrent spatial pattern

- **Recurrent spatial patterns** refer to the characteristic ways in which meteorological and climatological variables exhibit repeated, geographically organized behaviors over time. (e.g. El Niño–Southern Oscillation, ENSO)

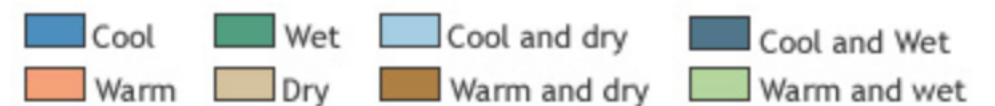
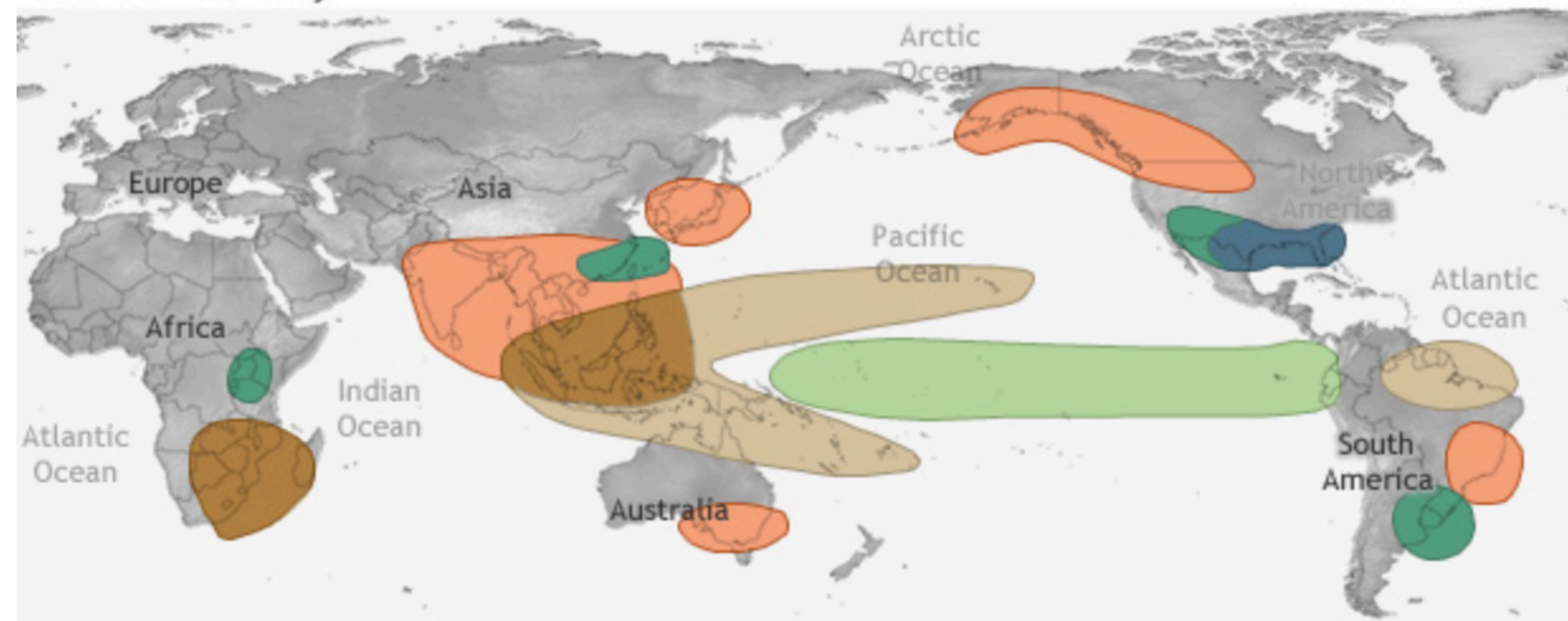


Recurrent spatial pattern

- **Recurrent spatial patterns** refer to the characteristic ways in which meteorological and climatological variables exhibit repeated, geographically organized behaviors over time. (e.g. El Niño–Southern Oscillation, ENSO)

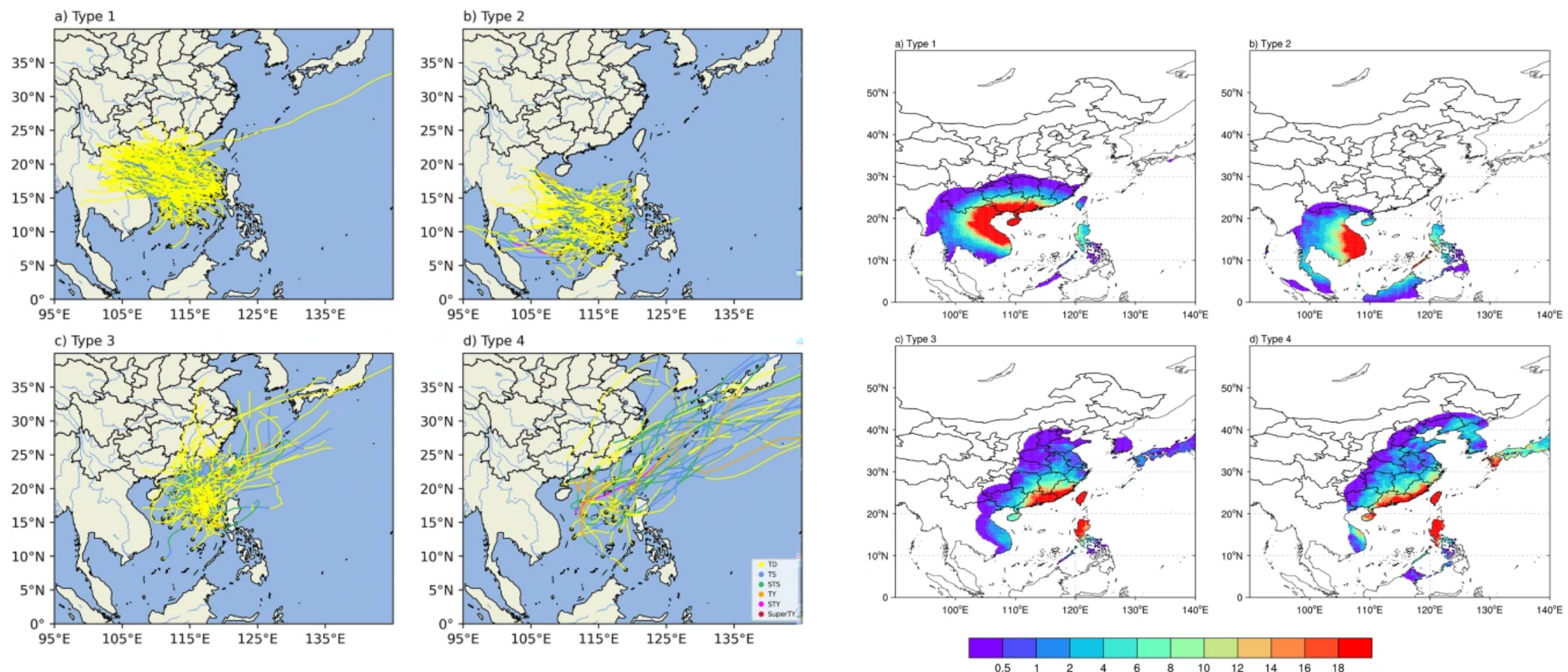
EL NIÑO CLIMATE IMPACTS

December–February



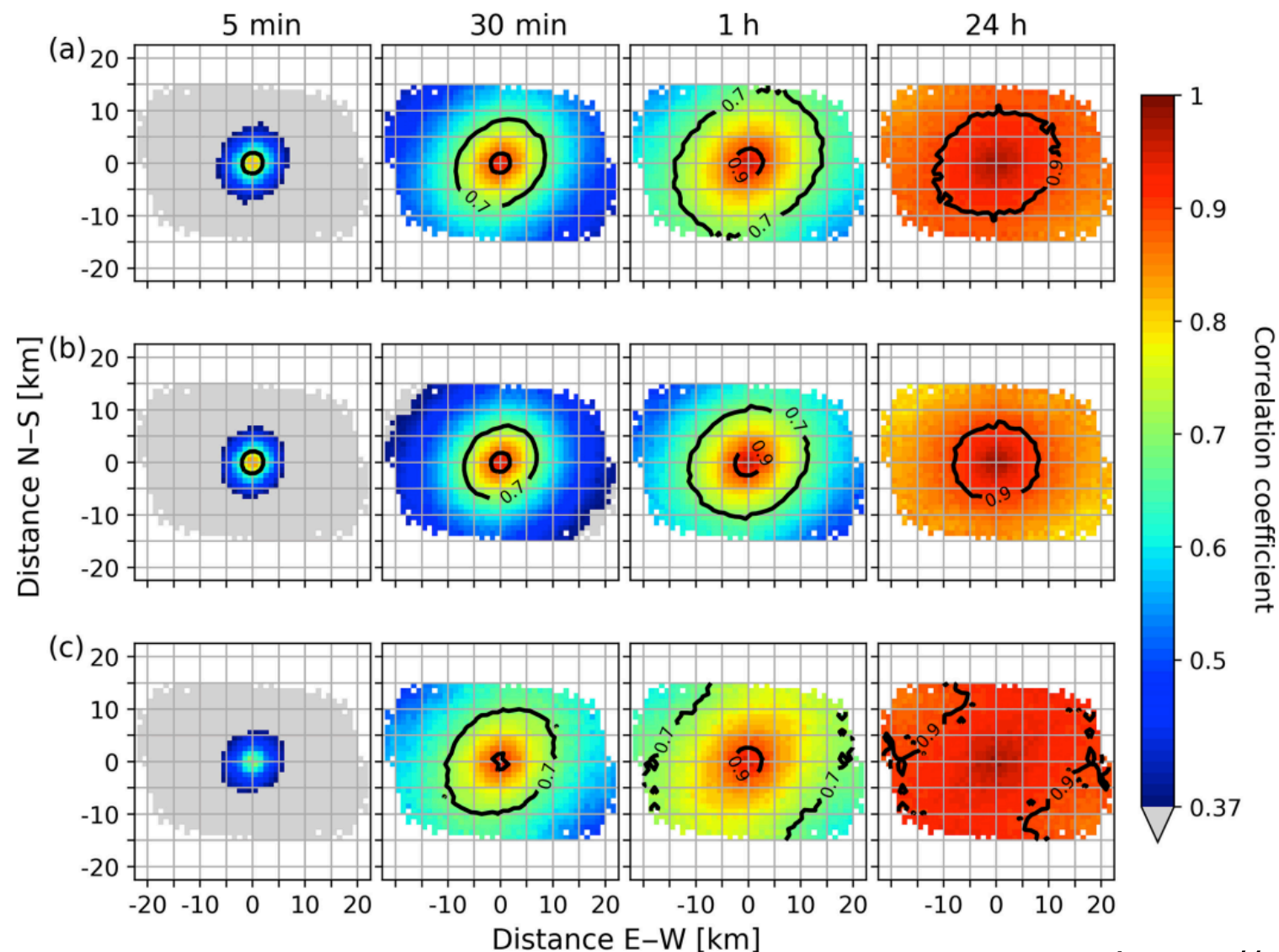
Spatial Clustering

- **Spatial clustering** refers to the tendency of meteorological and climatological variables to exhibit **non-random, geographically grouped behaviors**. These clusters reveal **coherent spatial structures**, such as regions that experience similar temperature trends, precipitation anomalies, or extreme events.



Spatial Autocorrelation

- **Spatial autocorrelation** refers to the degree to which nearby or neighboring locations exhibit similar values of a variable.



Weather and Climate Data



Weather is your mood, climate is your personality.

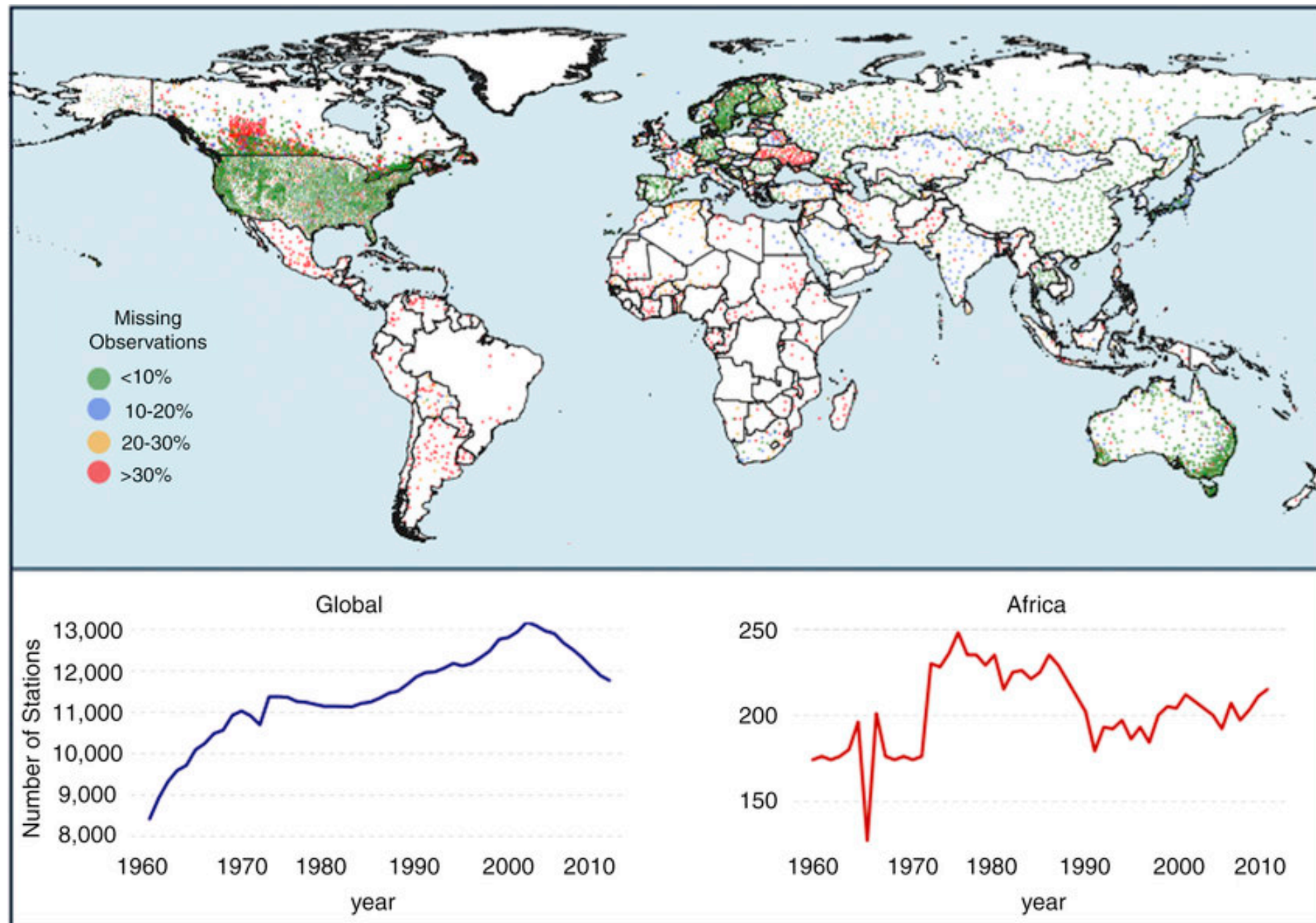
- Time series analysis: Weather focuses on short-term prediction, while climate targets long-term trends.
- Spatial analysis: Weather captures local phenomena, whereas climate reveals regional or global patterns.

<https://climatekids.nasa.gov/climate-change-meaning/>

Common data issues

- **Missing values** due to sensor gaps or transmission errors
- **Outliers** from faulty instruments or real extreme events
- **Inconsistent formats** (e.g., units, time, file types vary by source)
- **Large data volumes** from high-res global models or satellites
- **Resolution mismatch** between datasets in time and space
- **Others:** time zone mismatch, leap year issues, version updates

Missing data - ground obs



Missing data - satellites

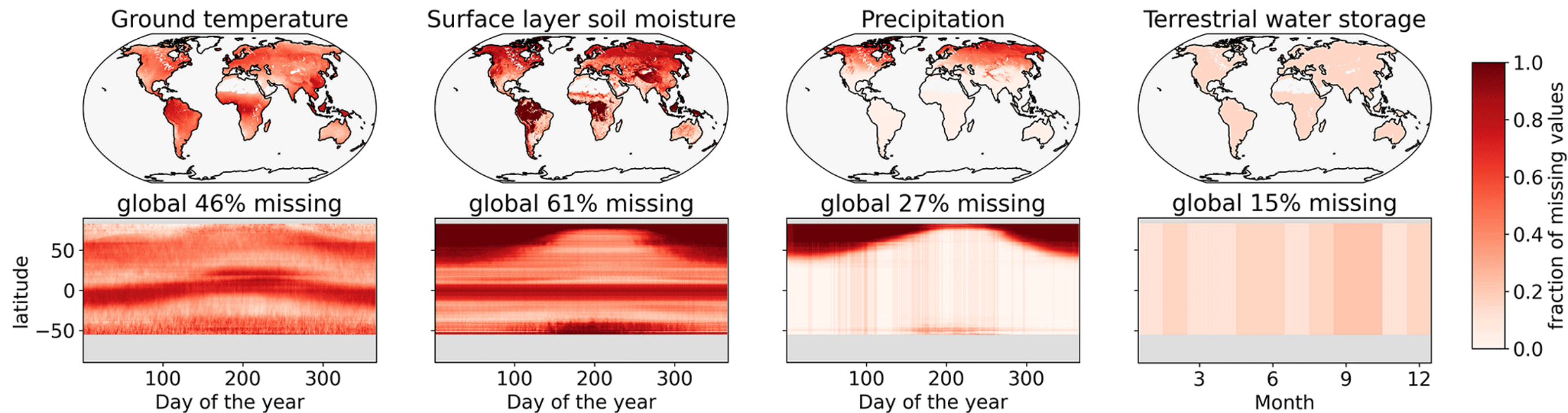
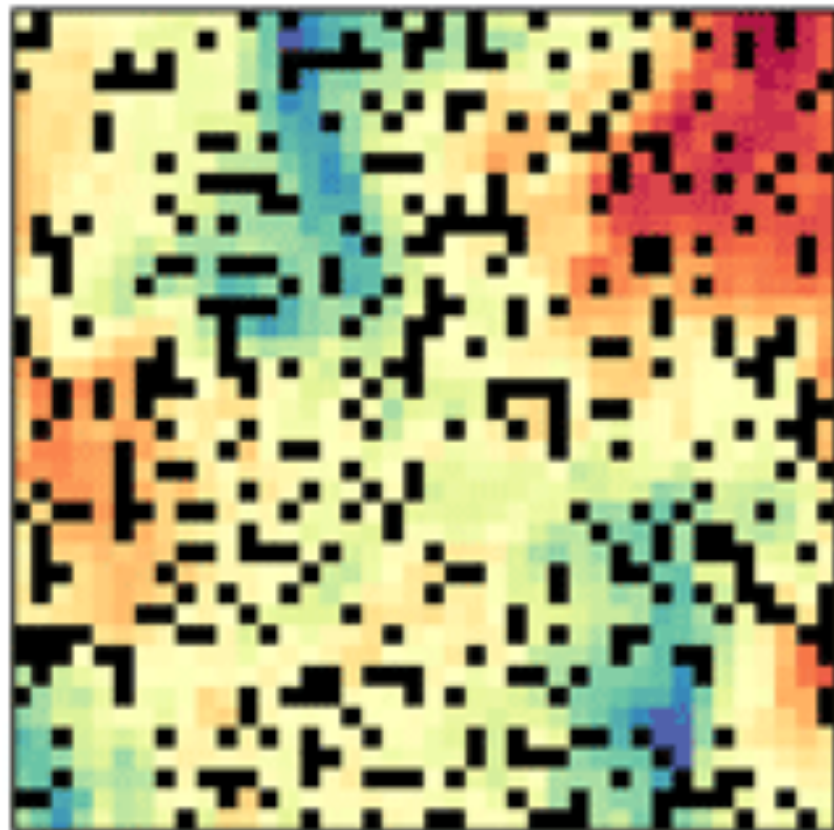


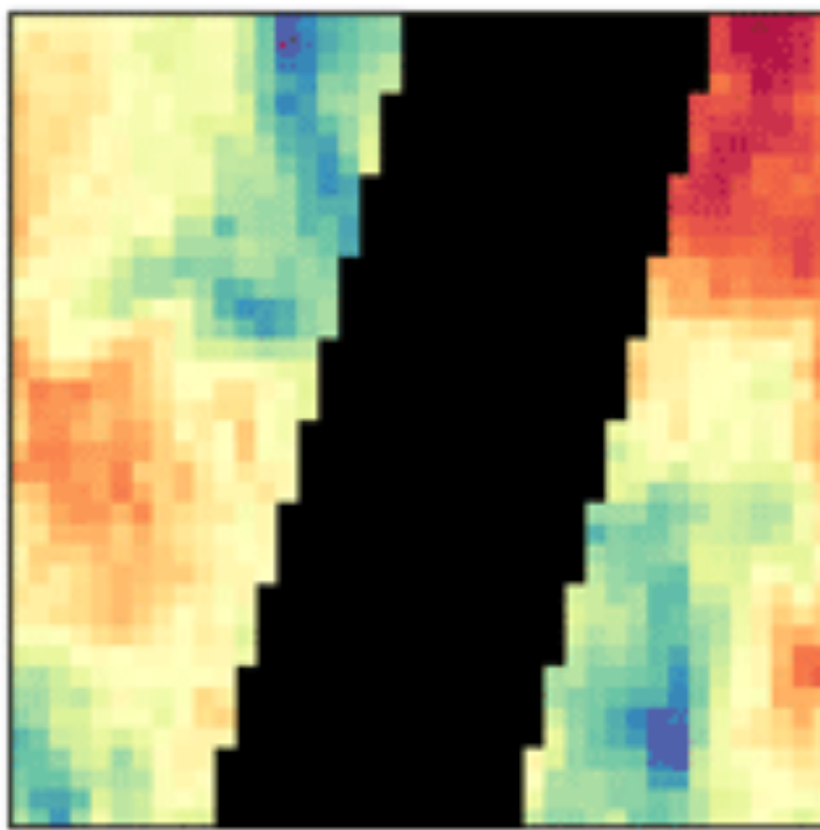
Figure 5 Fraction of missing data regarding ground temperature from MODIS, ESA-CCI soil moisture, GPM precipitation, and GRACE terrestrial water storage observations in the years 2003–2020. The upper panels show the fraction of missing data per land point on the ERA-5 grid, and the lower panels show fraction of missing values per latitude and day of the year. The data are down-sampled to daily values, with the exception of GRACE, which has monthly resolution.

Missing data

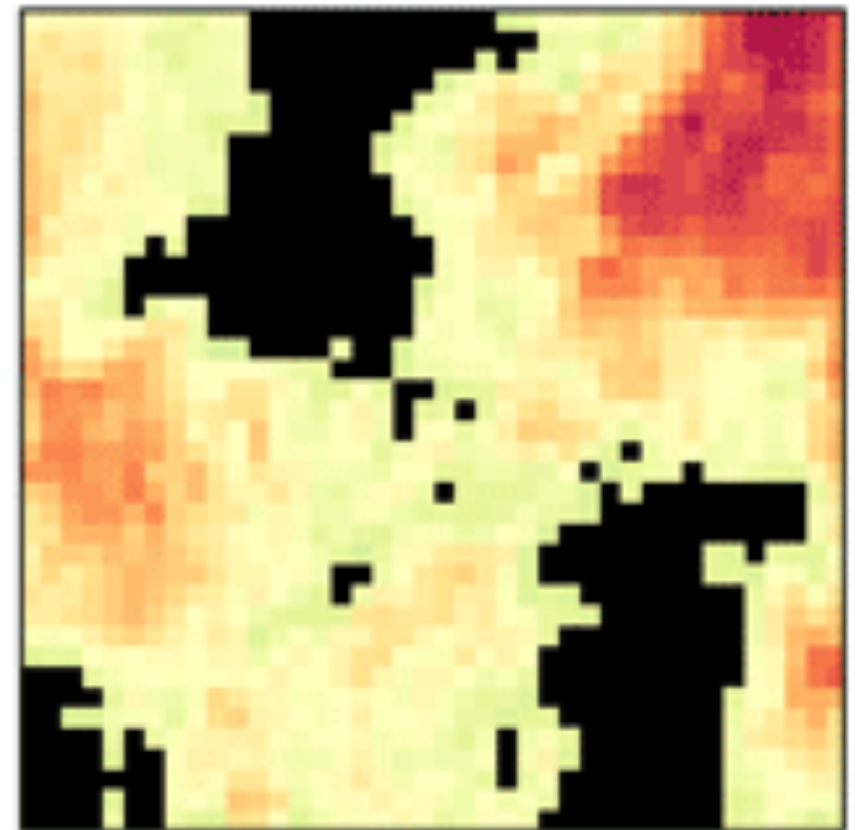
(a) Missing completely at random (MCAR)



(b) Missing at random (MAR)



(c) Missing not at random (MNAR)

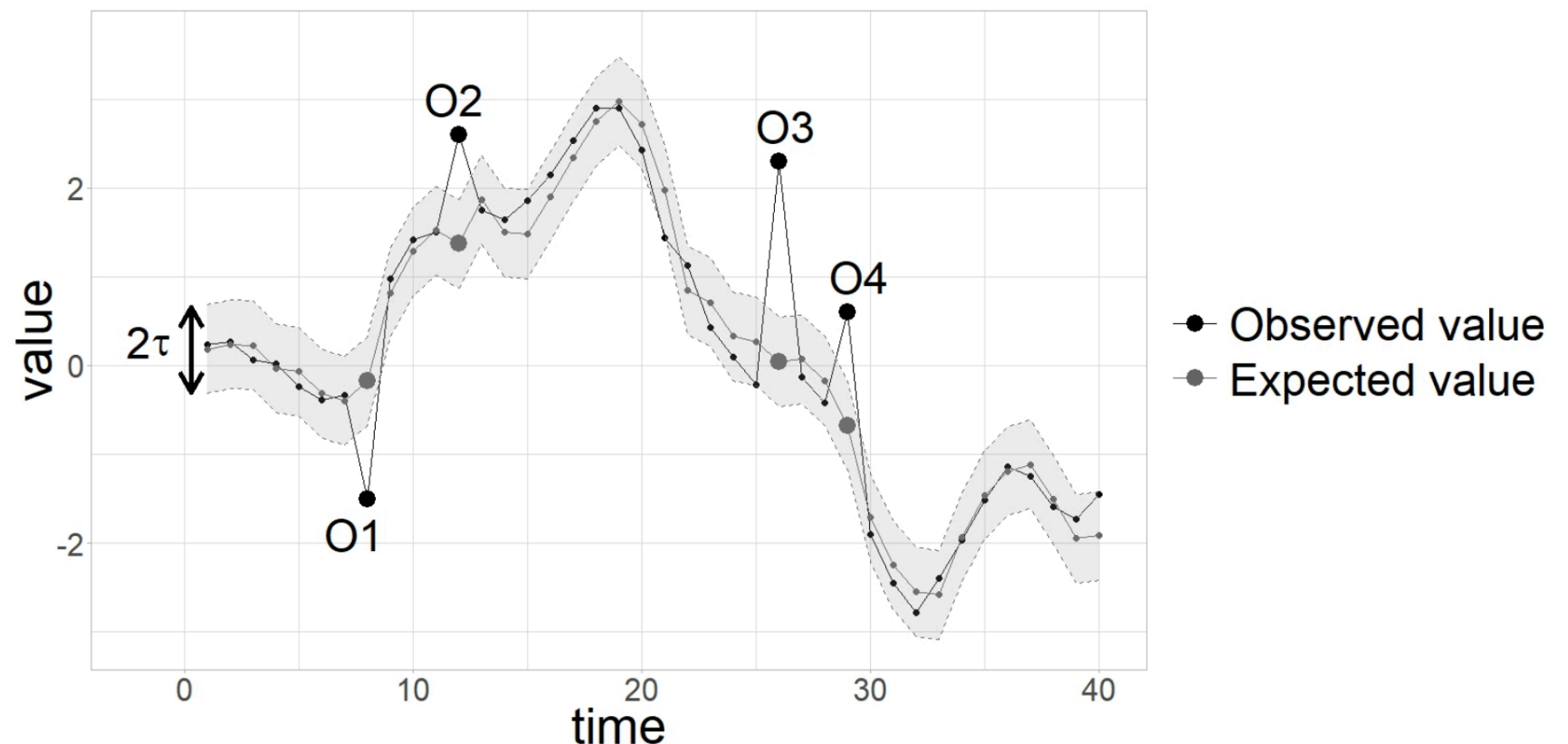


Missing data

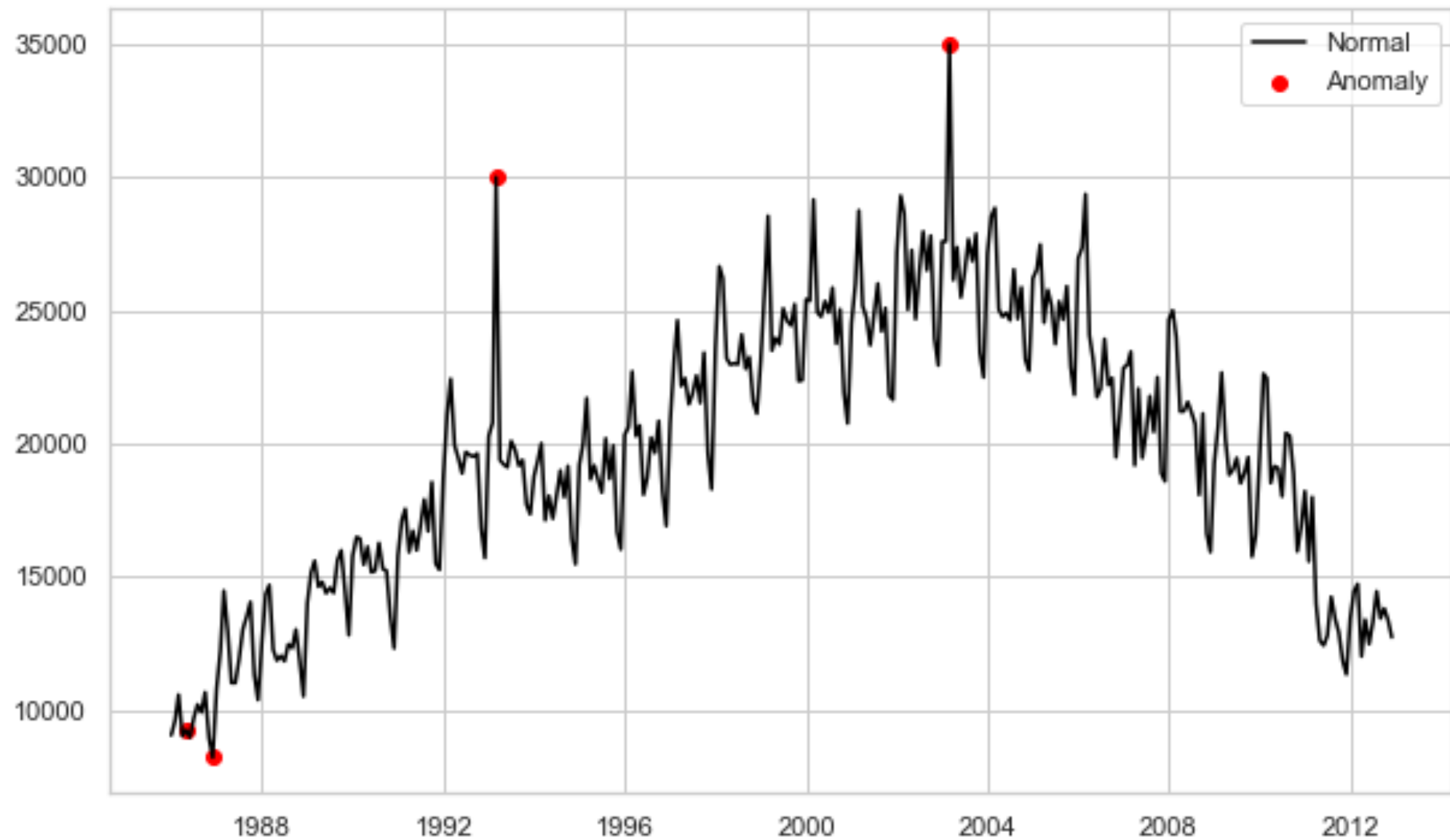
- **How to detect?**
 - Look for NaN, empty cells, or placeholder values (e.g., -9999)
 - Use `.isnull()`, `.notnull()` in pandas or masking in xarray
 - Visualize with time series plots (gaps in lines)
- **How to handle?**
 - **Small gaps:** linear interpolation, forward/backward fill
 - **Larger gaps:** mean/median imputation or model-based filling
 - **Sometimes:** just drop rows or columns if proportion is low

Ourliers

- Outlier (Hawkins, 1980): “An observation which deviates so much from other observations as to arouse suspicions that it was generated by a different mechanism.”
- Unlike extreme events, which are valid and physically meaningful, outliers may result from errors or noise and often require scrutiny or removal.

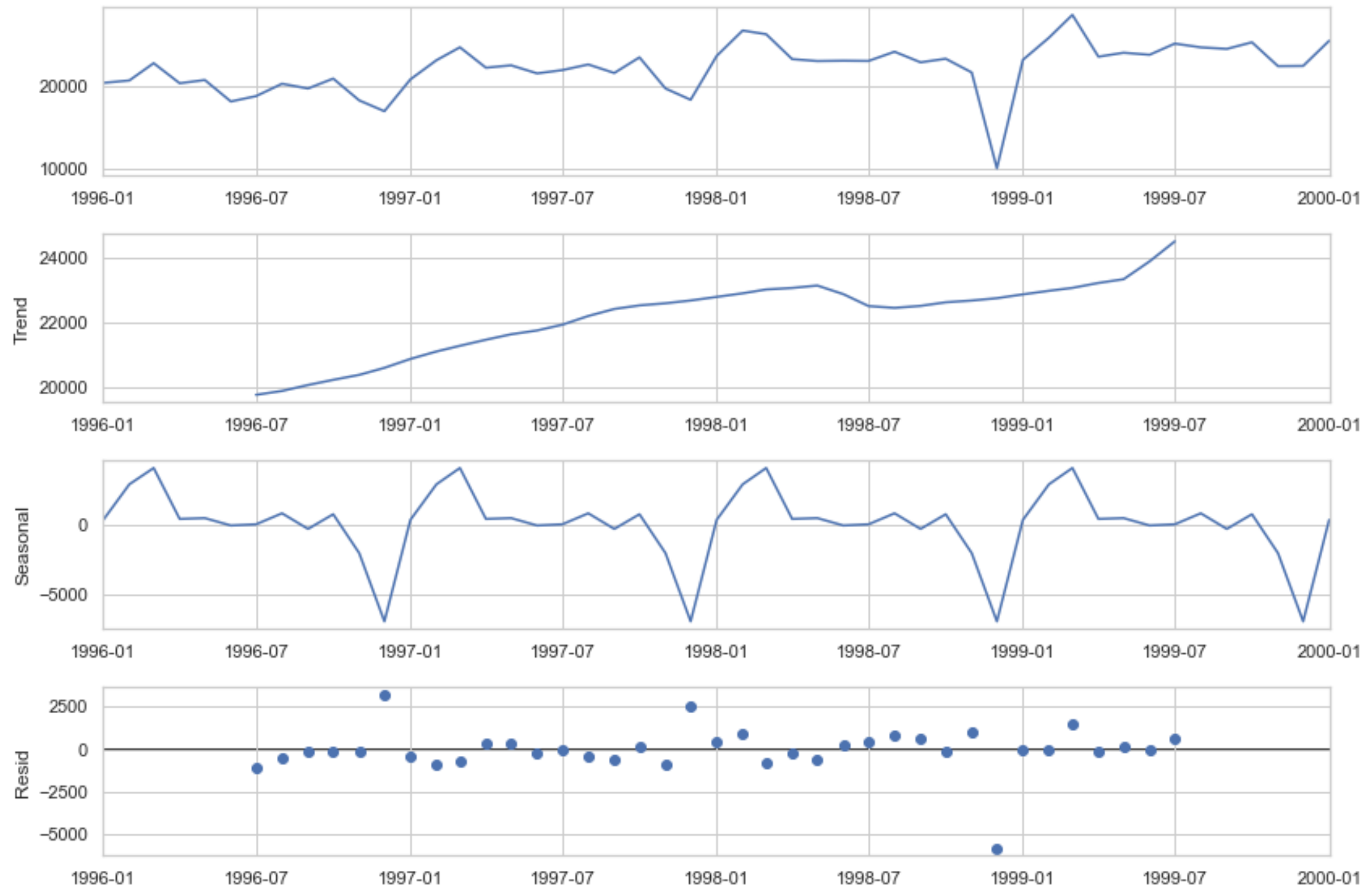


Ourliers



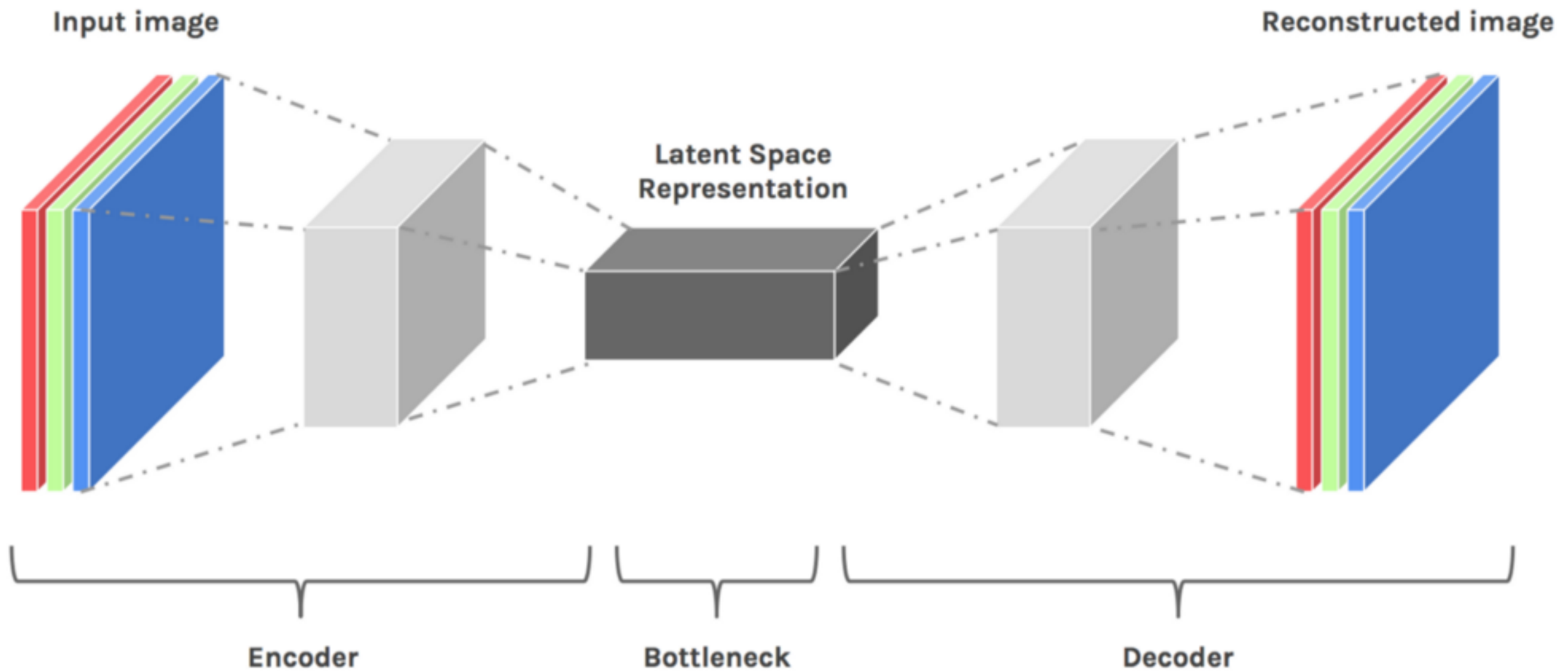
<https://neptune.ai/blog/anomaly-detection-in-time-series>

Ourliers



<https://neptune.ai/blog/anomaly-detection-in-time-series>

Ourliers



Inconsistent formats

- **Inconsistent formats** refer to differences in how data is stored, labeled, or structured — even when describing the same variable.

Inconsistent formats

- **Inconsistent formats** refer to differences in how data is stored, labeled, or structured — even when describing the same variable.
- **Units:**
 - Temperature in °C vs. K
 - Precipitation in mm/hr vs. mm/day
- **Time formats:**
 - UTC vs. local time
 - Missing time zones or inconsistent daylight saving adjustment
- **File structures:**
 - CSV, NetCDF, GRIB — different formats require different tools
 - Column names, headers, or time stamps may not align